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"Federated Learning Architecture for Health Monitoring and Anomaly Detection in Multi Satellite Systems"

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UE23CS320B – Capstone Project Phase - 2

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CERTIFICATE

This is to certify that the dissertation entitled

'Federated Learning Architecture for Health Monitoring and Anomaly Detection in Multi Satellite Systems'

is a Bonafide work carried out by

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In partial fulfilment for the completion of sixth-semester Capstone Project Phase - 2 (UE23CS320b) in the Program of Study -Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period Jan. 2026 – May. 2025. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 6th-semester academic requirements in respect of project work.

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DECLARATION

We hereby declare that the Capstone Project Phase - 2 entitled **"Federated Learning Architecture for Health Monitoring and Anomaly Detection in Multi Satellite Systems"** has been carried out by us under the guidance of **Dr. Richa Sharma, Associate Professor** and submitted in partial fulfilment of the course requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester Jan – May 2026. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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ABSTRACT

A new challenge within aerospace engineering is how to continuously monitor the condition of multiple different interconnected satellite systems simultaneously, in real time due to limitations imposed by bandwidth and latency constraints of a single grounded-based telemetry processing system. This paper describes the design, architecture and current state of development of a federated learning (FL) framework that supports autonomous health monitoring and anomaly detection in a simulated multi-satellite system.

The proposed framework operates off the ESA Anomaly detection benchmark (ESA-ADB) Mission 1 dataset which includes 76 disparate telemetry channels sampled at non-uniform rates including sub-second bursts to 15minute intervals. Each simulated satellite node will locally only process a subset of its assigned channels (i.e., a single satellite node will only recieve telemetry data from a subset of the telemetry channels on that satellite). Each satellite node will build an onboard DL model to reconstruct normal cyclic sensor behaviour based on the simulated channel data it has collected and then determine whether the reconstructed normal behaviour differs from what was measured as a deviation (i.e. identify anomalies). Each satellite node will only send a compressed set of model weight updates (to be used by other satellites in the constellation) to a rotating leader satellite node. The rotating leader satellite will receive the compressed model weight updates from all of the satellites in the constellation, perform a robust trimmed mean aggregation of the received updates to eliminate contributions from faulty nodes, and use those to build a globally optimised deep learning model for the satellite constellation as a whole. This new system allows for the elimination of the need for raw telemetry from the satellite to the ground. It targets a reduction of 90% in downlink bandwidth while maintaining a high level of anomaly detection accuracy.

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Compared to traditional space missions, newer missions utilize coordinated satellites swarms opposed to a single monolithic mission embodiment. US NASA's starling mission (4 satellites) and the ESA constellation (3 satellites for earth-based observation) demonstrate that distributing the mission functions throughout multiple nodes enhances system redundancy and coverage as well as better overall resilience. NASA is planning various deep-space exploration missions that will require swarm-based hardware composed of between 10 to 100 satellites working together. Commercial satellites operators are deploying multiple existing Low Earth Orbit constellations at unprecedented scale.

An Individual satellite in such swarms produces constant telemetry data streams from multiple subsystem sensors like: Electrical Power System (EPS), Thermal Control System (TCS) etc. Previously monitoring the health of each individual satellite has required transmitting raw telemetry data to the ground station and performing centralized analysis. Transmitting raw telemetry data to the ground station for centralized analysis faces two main limitations: limited bandwidth and long reaction times. As the sizes of swarm satellites increases, the aggregated size of the telemetry data being produced by the satellites exceeds the total amount of available downlink capacity and the delay for a ground-station based fault response is inconsistent with the time-critical need to protect the hardware.

Federated Learning (FL) provides a principled approach to both the issues. FL allows for the development of local anomaly detection models for each individual satellite in the swarm using local telemetry data collected from that individual satellite and performing those calculations without having to transmit any raw telemetry data back to the ground station.

1.2 Project Overview

The goal of this project is to develop and implement a decentralized federated learning framework for monitoring the health of a multi-satellite system and for detecting anomalies. This framework will operate on ESA-ADB Mission-1 dataset which provides realistic telemetry dataset from an actual ESA satellite. The ESA-ADB Mission-1 dataset has 76 telemetry channels with complex inter-sensor dependencies; it has annotated windows that describe anomalies (i.e. true faults) from rare nominal events; and it has a record of telecommands for changing the operational state of the satellite.

The multi-satellite system will be simulated as a swarm of four satellites where the 76 telemetry channels are partitioned across four virtual satellite nodes. Each virtual satellite node will then train a local anomaly detection model on a subset of the 76 channels assigned to its corresponding virtual satellite node. The virtual satellite node with the highest model accuracy will serve as the rotating leader of the satellite swarm and will aggregate model updates from all nodes using a core robust statistical filter. Further to pre-emptively notify satellites operators of impending subsystem failures before they become catastrophic, the virtual satellite swarm is augmented with a predictive ground alerting system. The swarm architecture provides for scalability, fault-tolerance, bandwidth-efficiency to directly support the operational constraints of real satellite swarms.

1.3 Scope of Phase-2

This document shows all the accomplishments made between January and May 2026 as part of the second stage of the Capstone Project. The second stage has three primary deliverables:

(1) A full data preprocessing pipeline consisting of both telemetry resampling and feature engineering.

(2) A completed design for an entire system architecture as well as a comprehensive documentation for each class's design.

(3) Initial implementation of a local anomaly detection model & federated learning communications framework.

Additionally master datasets will have both annotated set of anomalies and an annotated set of telecommands prior to this project being concluded.

1.4 Report Organization

This report contains the following sections. Chapter 2 describes the problem statement and describes the technical issues that were resolved. Chapter 3 describes the ESA-ADB data set, provides a detailed statistical analysis of the 76 telemetry channels, and the complete methodology for data preprocessing was carried out. The proposed design for the proposed system is presented in Chapter 4, including the novelty, innovation, reliability, security, and all other software quality attributes. The high-level architecture of the system is outlined in a three-tiered structure (Chapter 5). Chapter 6 contains detailed design documents including class diagrams, ER diagrams, use case diagrams, and external interface diagrams. Chapter 7 lists technologies and technologies used. Implementation status and pseudocode are provided in Chapter 8. Chapter 9 summarizes Phase Two of this project. In Chapter 10, Phase Three will be described, along with a timeline for its completion.

CHAPTER 2

PROBLEM STATEMENT

Centralized ground-based monitoring of the satellite swarms in the complex orbital setting suffers from extreme bandwidth saturation and latency in response to the high dimensionality of telemetry streams from the many different assets being monitored as they move through space simultaneously. Each satellite continuously generates readings from multiple channels generated by onboard sensors associated with power, thermal, propulsion, attitude determination, and data handling subsystems (for example, telemetry data, temperature, and actual state from each of these subsystems). Collecting this raw telemetry for analysis at ground-based facilities using bandwidth over a large number of satellites is impractical from a bandwidth perspective; in addition, the time required for signals to travel to the ground from the satellite limits the likelihood of having real-time access to hardware faults that are evolving quickly.

There have not been any existing frameworks for detection of anomalies that have the capability to enable decentralized collaborative autonomy, which would allow peer-to-peer learning between onboard satellites; statistically filtering out the contributions of faulty nodes to a determined anomaly; and alerting ground personnel of a potential problem with a satellite during the critical hardware degradation time frame. When a satellite in a swarm exhibits an early indication of subsystem failure, there is no way for it to share that knowledge with its peer satellites that may also be showing symptoms of failing subsystems. As a result, all satellites are learning in isolation, and therefore never realize the full potential of the collective intelligence of the swarm.

This project will address the following specific technical challenge problems:

(1) How do you partition heterogeneous telemetry channels across satellite nodes in a principled manner that maintains dependencies between the satellites?

(2) How do you develop local models that enable anomaly detection for on-board satellites and provide real-time feedback to all members of the swarm?

2.1 Existing limitations

The current architecture used for monitoring the health of satellites has the capability to transmit raw telemetry to ground stations for analysis by a human operator or centralized software on the earth. However, this architecture has three main limitations:

- It does not scale well because the telemetry volume increases linearly with the increase of constellation size, but the available bandwidth for downlinking data does not increase accordingly.
- It has a large latency due to the time it takes for an anomaly to be detected by the ground, and for commands to be sent back to the satellite to correct for the anomaly.

Although there has been some research into using on-board machine-learning methods for detecting anomalies in satellites, this research has largely isolated training models for each satellite without a mechanism for sharing knowledge across multiple satellites. Federated learning has been used to simulate federated learning by splitting the time of the data associated with an anomaly over the various federated nodes. This is not an accurate representation of how a group of satellites operates in that a group of satellites will each only have access to their own sensor data. Therefore, each satellite will not have access to the collective knowledge of the different anomaly types that exist between the different satellites in the swarm. Additionally, there was no differentiation between an actual anomaly or rare nominal events; therefore, the application of the algorithm will report a higher true anomaly rate that will negatively impact the overall precision of the anomaly detection.

2.2 Proposed Solution

A Federated Learning Architecture is designed to solve all these problems. Every saturated node creates its anomaly detection model on its unique channel. Then it creates a health index to reflect the health of each subsystem. After each node creates its health index, it only sends compressed model weights to a rotating leader. The leader uses Trimmed Mean aggregation to remove faulty node's contributions, builds a new set of global models for each node and returns them back to that node. This operation continues to run autonomously without the need for ground support and the transmitted model weights are greatly smaller than the comparable raw telemetry streams, typically by orders of magnitude.

This architecture has a 3category labelling system that allows for true anomalies to be separated from the rare nominal events and gaps in the communications. There is also a predictive Remaining Useful Life (RUL) estimator that will take linear regression from historical health index trends to determine when subsystems are going to fail many hours or days in advance which will trigger emergency notifications to ground stations for emergency prevention before it is too late. Also, there is an autonomous failure detection/identification/recovery (FDIR) subsystem that can execute hardcoded safe mode protocols when subsystems cross allowable degradation thresholds to provide hardware protection as a last resort without requiring ground contact.

CHAPTER 3

DATA

3.1 Overview

The choice of dataset is foundational to any machine learning system. For a project addressing satellite telemetry anomaly detection, the dataset must provide realistic multi-channel telemetry with genuine annotated anomalies, sufficient temporal extent to capture long-duration fault progressions, and documentation of external events (such as ground-commanded operational changes) that may produce sensor signatures resembling anomalies but are not. The European Space Agency Anomaly Detection Benchmark (ESA-ADB) satisfies all these requirements and is the dataset of choice for this project.

3.2 Dataset Description

3.2.1 ESA Anomaly Detection Benchmark – Mission 1

The ESA Anomaly Detection Benchmark (ESA-ADB), developed by Gerhard, Boin, and Fougeres (2024), is the most thorough publicly available satellite telemetry anomaly detection benchmark dataset. The dataset comes from the actual operational telemetry of ESA spacecraft and contains datasets from multiple missions of different complexities. This project will use Mission 1 dataset, which consists of 76 telemetry channels and all operational telemetric data like the Telecommands with the full observation window spanning approximately 15 years. The 76 channels are organized into 18 groups based on physical subsystem similarity, covering the EPS (Electrical Power System), TCS (Thermal Control System), GNC (Guidance, Navigation and Control), ADCS (Attitude Determination and Control System), propulsion, data handling, and payload subsystems. The dataset also comes with per channel anomaly annotations to indicate the precise start time and end time of each anomaly occurrence, along

with a categorisation of whether each occurrence was a true anomaly (genuine malfunction), rare nominal event, or a communication gap (loss or poor quality of telecommunication).

3.2.2 Target and Non-Target Channels

The 76 target channels for anomaly detection are appropriately distinguished into 2 groups: Target and non-target channels. There are 58 target channels in Mission 1 and 18 nontarget channels. The nontarget channels are used as context for detecting anomalies, not providing any pertinent information for global design decisions. Only the target channels are used for anomaly detection.

The determination of resampling rate and the appropriate window size must be based solely on samples from the target channels (i.e., no loss of fidelity on any anomaly events occurring in target channels), while samples from non-target channels will be temporally aligned with (via resampling) the target channel grid; however, the individual sampling characteristics of non-target channels should not affect the global design choices.

3.2.3 Anomaly Annotation

The Mission 1 data set contains 200 event annotations for the entire range of the collection, with 118 of them being identified as genuine anomalies. The remaining 78 are identified as rare nominal events that are unusual but, in this case, are intentional and created by commands sent to the spacecraft via telecommand signals. The final four event annotations are identified as having gaps in communication (in this case, interruptions or delays in transmission of telemetry data).

The time duration for each of the anomalies/data points varies from point anomalies (with a time duration of zero seconds) that require snap to nearest grid slot type handling to multi-week (over 1.2 million seconds) events.

Point anomalies (11 in total) require special handling for label annotation. Since point anomalies cannot be assigned to 30-second grid slots using the regular method of assigning time to grid slots using intervals (i.e., marking the interval where the beginning of each event would be), they must instead be assigned to the single nearest (closest) 30-second grid slot before their recorded timestamp.

For example, when there is an extended duration of an event, such as the event identified as having a duration of 612,810 seconds, this event will map to 20,427 consecutive 30-second grid slots, each marked with the appropriate event label as recorded in this data set.

3.2.4 Telecommand Records

There are a total of 698 telecommands issued in Mission 1 of the ESA-ADB dataset, of which 668 of them have priority 0 or 1. These telecommands perform routine operations but do not affect the state of the satellite. The remaining 30 telecommands are priority 2 (19 commands) or priority 3 (11 commands) and represent significant operational state changes that are directly associated with the 78 rare nominal events in the anomaly annotations. These 30 high priority telecommands are included in the models input features.

The data contain (698) all the telecommands issued to the satellite during the observation period (30 seconds); however, the first (668) priority "0 or 1" are assigned as routine telecommands and will create an insignificant impact when determining the signatures as described previously. The other (30) telecommands, (19) assigned priority "2" and (11) assigned priority "3" will create significant impacts on the operational state of the satellite and are associated with the other (78) nominally rare events identified in the anomaly annotation. The (30) high priority telecommands will be used in the final feature set for the input to the model, as a binary indicator column on the 30-second frequency.

3.3 Data Preprocessing

3.3.1 Channel Sampling Rate Analysis

The Mission 1 dataset consists of 76 channels, each of which samples data at very different, non-uniform rates. To establish a proper preprocessing pipeline, a thorough and detailed statistical analysis of the sample intervals between all 76 channels provided useful information. The following statistics were calculated for each of the channels: average, median, mode, and standard deviation of the inter-sample intervals; the dominant interval (the interval that occurs the most); the percentage of the dominant interval of all sample intervals; and the top-3 most frequently occurring intervals and their respective percentage values.

Based on this analysis, we have determined that these 76 channels can be categorized into 2 fundamentally different behavioural classifications. One category contains channels that primarily sample at a highly predictable rate or have a very stable and consistent sampling pattern (known as Zero Order Hold [ZOH]). The characteristics of these types of channels are: 1) the dominant interval is greater than or equal to 30 seconds; and 2) the percentage of the most dominant interval is 90% or higher.

The second category has variable sampling behaviours (known as aggregate channels) and consists of channels that have at least 1 of the following characteristics: 1) the dominant interval is less than 30 seconds; or 2) the percentage of the dominant interval is less than 90% of the sample intervals. Channels exhibiting aggregate characteristics typically sample data at a bimodal or highly variable rate and as such require statistical feature extraction as opposed to simple resampling of the data.

3.3.2 Channel Classification Results

The analysis classified 44 of the 76 channels as ZOH channels and 32 channels as aggregate channels. The aggregate channels are organized into distinct behavioural groups based on their dominant interval patterns:

- Group 1 (channels 1–3): Bimodal distribution between 3-second and 90-second intervals, with 3 seconds as dominant at 51.65% occurrence. These are non-target context channels.
- Group 2 (channels 4–11): Bimodal distribution between 0.75-second and 180-second intervals, with 180 seconds dominant at 45.48% occurrence. Non-target context channels.
- Groups 3 and 5 (channels 12–13, 17–20, 25–28, 34–37): Sub-second dominant interval of 0.38 seconds at 58.41% occurrence, with 90-second polling mode at 33.82%. These are target channels with approximately 79 raw readings per 30-second window during high-frequency mode.
- Groups 15–17 (channels 64–69) and channel 74: Bimodal distributions between 3-second and 180-second intervals, or 30-second dominant with below-threshold dominance percentage (70.83% for channel 74).

The ZOH channels include all channels whose dominant interval is 30 seconds or 90 seconds with dominance percentages above 90%, including the critical target channels 41–46 (dominant interval 30 seconds, 90.28% dominance), channels 47–52 (dominant interval 30 seconds, 99.81% dominance), and channels 53–60 (dominant interval 30 seconds, extremely clean pattern).

3.3.3 Resampling Strategy

All 76 total channels have been transformed into 30 second windows of data based upon their original timeframes. Each of these blocks are related by a common start time of 2000-01-01 00:00:00 UTC. The following list explains why 30 seconds was chosen for the conversion from a time series to an equal valued time series:

1. The dominant sampling rate of critical channels (41 though 53) matched this 30 second period.
2. ESA-ADB benchmark article utilized this frequency in its conversion for Mission 1.
3. Real detail was retained from fast sampled channels (sub half second) and slow sampled channels (over three-minute intervals) were not over sampled.

For ZOH channels, the conversion method utilized by the model is zero-order hold forward filled with each sensor value filling the next 30-second slot until the next measurement was received thereby reproducing the actual state of the satellite's sensor based upon its most current physical reading. The physical reading on the site of a satellite physical sensor is effectively maintained until the next conversion occurs. Slow sampled channels, with a 90-second relative dominant period, will produce a repetitive step function in the converted signal with three identical numbers followed by an instantaneous transition to the next value, which will be the normal identification of these channel types by the model.

For aggregate channels, during each 30-second window of time, resampling generates five fixed statistic features, which are the following: mean (mean average of all raw readings during the time period), max (maximum value), min (minimum value), std (standard deviation), and count (the number of real physical readings for that channel during the time period). The count feature is particularly useful for identifying anomalies because it tells us whether the channel was in high-speed mode or in slow polling mode for each window. If a channel that is typically in high-speed mode suddenly has count=0 for an extended period, then that channel is exhibiting a change in operating mode that will be flagged by the model.

For aggregate channels during windows that do not receive any raw readings (i.e., polling mode windows that fall between high speed bursts of data), mean, max, and min are filled forward from the last valid real reading; however, std is set to 0.0 and count is set to 0.

3.3.4 Label Annotation

Label annotation is currently underway and is the next step in the preprocessing pipeline. Three binary label columns are used in the master 30-second grid: is_anomaly (set to 1 for all 30-second slots falling within a true anomaly window), is_rare_event (set to 1 for all slots within a rare nominal event window), and is_comm_gap (set to 1 for all slots within a communication gap window). The three columns are independent and may both be 1 simultaneously if events overlap, making this a multi-label binary annotation scheme.

Point anomalies (11 events with duration_seconds = 0) require special handling: they are snapped to the nearest 30-second grid slot preceding their recorded timestamp, and only that single slot receives a label of 1. All other anomaly events are stamped continuously across all grid slots from their start to end timestamp. The resulting labels_master.csv and telecommands_master.csv files will be merged with the feature matrix on the datetime index to produce the final master_dataset.csv that will be used for model training.

3.3.5 Train/Validation/Test Split

The dataset is split chronologically into training, validation, and test sets. Random splitting is explicitly avoided because it would allow information from anomaly periods to leak into the training set. The split has been made in accordance with the ESA Benchmark article, where the authors have mentioned the optimal and the recommended data split that must be followed.

Supplementary Table 11. **Statistics of training, validation, and test sets for different phases of Mission1 considering the full set (top panel) and the lightweight subset of channels (bottom panel).**

Mission1 – the lightweight subset	Phase 1		Phase 2		Phase 3		Phase 4		Phase 5		**Test**
	Train	Val	Train	Val	Train	Val	Train	Val	Train	Val	
Data points	**1,125,600**	**314,399**	**3,900,977**	**997,530**	**8,900,105**	**1,479,360**	**19,171,279**	**1,463,274**	**39,774,080**	**1,479,370**	**40,925,288**
Telecommands' executions	7,769	15,918	94,426	45,194	271,882	13,295	414,927	9,001	764,648	60,157	769,917
Duration (anonymised)	9 weeks	3 weeks	8 months	2 months	18 months	3 months	39 months	3 months	81 months	3 months	84 months
Annotated points [%]	1.41	17.29	2.76	1.49	3.24	0.02	1.84	0.11	1.74	1.23	1.81
Annotated events	**1**	**1**	**6**	**3**	**17**	**2**	**28**	**1**	**52**	**3**	**65**
Anomalies	0	1	3	1	5	0	10	0	22	2	29
Rare nominal events	1	0	3	2	9	2	14	1	26	1	36
Communication gaps	0	0	0	0	3	0	4	0	4	0	0
Univariate / Multivariate	0 / 1	0 / 1	0 / 6	0 / 3	0 / 14	0 / 2	0 / 24	0 / 1	0 / 48	0 / 3	1 / 64
Global / Local	1 / 0	1 / 0	4 / 2	2 / 1	11 / 3	1 / 1	18 / 6	1 / 0	39 / 9	3 / 0	40 / 25
Point / Subsequence	0 / 1	0 / 1	0 / 6	0 / 3	0 / 14	0 / 2	0 / 24	0 / 1	1 / 47	2 / 1	9 / 56
Distinct event classes	**1**	**1**	**5**	**2**	**10**	**2**	**15**	**1**	**17**	**2**	**13**
Mission1 - the full set	Phase 1		Phase 2		Phase 3		Phase 4		Phase 5		**Test**
	Train	Val	Train	Val	Train	Val	Train	Val	Train	Val	
Data points	**8,954,221**	**2,176,171**	**29,416,435**	**7,890,008**	**68,888,013**	**10,761,293**	**144,775,815**	**10,273,971**	**305,515,601**	**10,741,556**	**428,599,738**
Annotated points [%]	2.11	10.62	1.87	1.52	1.96	0.93	1.32	0.03	1.33	1.62	2.25
Annotated events	**5**	**4**	**20**	**8**	**54**	**4**	**73**	**1**	**104**	**5**	**91**
Anomalies	4	1	13	2	27	2	40	0	59	4	55
Rare nominal events	1	3	7	6	24	2	29	1	41	1	36
Communication gaps	0	0	0	0	3	0	4	0	4	0	0
Univariate / Multivariate	3 / 2	3 / 1	11 / 9	5 / 3	31 / 20	0 / 4	31 / 38	0 / 1	31 / 69	0 / 5	1 / 90
Global / Local	3 / 2	4 / 0	11 / 9	5 / 3	36 / 15	1 / 3	44 / 25	1 / 0	67 / 33	3 / 2	43 / 48
Point / Subsequence	0 / 5	0 / 4	0 / 20	0 / 8	0 / 51	0 / 4	1 / 68	0 / 1	2 / 98	2 / 3	9 / 82
Distinct event classes	**3**	**2**	**6**	**3**	**11**	**3**	**16**	**1**	**18**	**3**	**17**

Figure 1: Dataset Train/Validation/Test Split according to the ESA-ADB dataset

CHAPTER 4

DESIGN DETAILS

4.1 Novelty

The primary novelty of this project lies in the combination of three independently researched ideas that have not been previously integrated: channel-partitioned cross-satellite federated learning, graph-based inter-sensor dependency modelling, and health-aware dynamic leader election. The existing FL approaches to satellite telemetry simulate federation by splitting time periods of a single satellite across multiple clients, which does not reflect the physical architecture of a real satellite swarm where each spacecraft has access only to its own on-board sensors. This project partitions channels across satellite nodes based upon subsystem grouping which is more challenging from a federated learning perspective because each node sees a fundamentally different feature space.

The integration of this combined GNN-FL approach to the ESA-ADB Mission 1 full 76-channel dataset is additionally a greater novelty as this most challenging public satellite telemetry has no published work addressed with GNN methods. Additionally, the preprocessing pipeline introduces several novel engineering contributions such as, systematic identification and special treatment of bimodal channels through a dual-strategy resampling approach, three-category annotation scheme that explicitly separates anomalies from rare nominal events and communication gaps, and the integration of telecommand context as auxiliary binary input features. While each of these elements appear individually in various research, their systematic combination on this dataset has not been demonstrated.

4.2 Innovativeness

The system introduces an innovative three-class label scheme that explicitly separates true anomalies from rare nominal events and communication gaps. This distinction is absent from all existing satellite telemetry papers and addresses a critical source of false positives: telecommand-induced state changes that produce unusual sensor signatures but are entirely intentional. By encoding high priority telecommands as a binary context feature and separating rare nominal events from true anomalies in the training labels, the model runs to associate certain unusual patterns with legitimate operational commands and substantially reducing false alarm rates.

A further innovation is the health-aware leader election protocol. Unlike standard federated learning architectures that use a fixed server for aggregation, this system elects the aggregator for dynamically based on the Health Index of each node. The satellite with the highest current health score becomes the aggregator for the next N rounds, ensuring that the most reliable node in the swarm is always responsible for weight aggregation. This provies a defense in depth against both failing satellites and model poisoning attacks combined with Trimmed Mean Robust Aggregation.

The project innovates at the intersection of graph-based representation learning and federated optimisation. The federated partitioning by channel group means that each satellite node's local model sees only a subset of the full inter-channel graph. This creates a novel technical challenge: how to aggregate graph-structured local models into a globally consistent model that captures cross-subsystem dependencies that no individual node can observe alone. FedProx prevents nodes with very different graph structures and loss landscapes from diverging during federation.

The telecommand encoding scheme also represents an innovative contribution: by encoding high priority telecommands as binary impulse features on the same grid as the telemetry, the model gains the ability to condition its anomaly score on known commanded operations. This allows it to learn the distinction

between commanded rare events (TC fired + signal change = expected) and genuine anomalies (no TC fired + same signal change = anomalous).

4.3 Interoperability

The architecture is designed to accommodate satellites with heterogeneous channel counts and feature dimensions: FedProx regularization handles the non-IID nature of the channel partitions by penalizing divergence from the global model, enabling productive aggregation even when individual satellites are monitoring very different subsystem types.

The preprocessing pipeline is designed to be fully interoperable with the ESA-ADB data format. Raw input is consumed from the standard ESA-AD folder structure, and all output files conform to a shared 30-second master grid that allows any downstream model to join feature, label, and telecommand data by timestamp index without additional alignment operations. The label and telecommand master CSV files can be consumed by any TSAD algorithm, not just the GNN-FL approach developed in this project.

The Flower federated learning framework used for simulation is interoperable with standard PyTorch model definitions, meaning the local GNN model implementation can be swapped for any alternative architecture (LSTM, Transformer, CNN) without changes to the federated training orchestration code.

4.4 Performance

Performance is evaluated using the ESA-ADB hierarchical evaluation pipeline, which assesses five priority aspects in order:

(1a) minimising false alarms through the corrected event-wise F0.5-score

(1b) anomaly existence recall

(2a) subsystem-aware F0.5-score

(2b) channel-aware F0.5-score

(3) alarming precision

(4) detection timing quality (ADTQC)

(5) anomaly range and proximity through the modified affiliation-based F0.5-score.

The corrected event-wise F0.5-score is the primary performance metric as it uses beta=0.5 to weight precision twice as heavily as recall, reflecting the operational reality that false alarms are more disruptive to mission control than missed detections. Target performance is to exceed the Telemanom-ESA-Pruned baseline of F0.5 = 0.786 on the Mission 1 lightweight subset (i.e., channels 41 to 46).

The system targets sub-500 millisecond latency for on-board anomaly detection and Health Index computation, meeting the real-time response required.

4.5 Security

The federated learning architecture provides inherent security benefits over centralised approaches. Raw telemetry data never leaves the simulated satellite node, only the model weight gradients are communicated to the aggregation server. This preserves data privacy and reduces cybersecurity attack surface. The Trimmed Mean aggregation strategy provides a form of Byzantine fault tolerance against model poisoning attacks. The leader node prevents any single satellite with compromised or malfunctioning model weights from corrupting the global model by discarding weight updates that deviate more than a z-score threshold from the swarm mean. The health-aware leader election protocol adds a second layer of security.i.e., a satellite exhibiting anomalous health characteristics is unlikely to be elected as the aggregator, limiting the influence of potentially compromised nodes on the federation. Since raw telemetry never leaves the individual satellite, the system also provides data privacy protection against interception of sensitive mission data during transmission.

4.6 Reliability

System reliability is ensured through a dual-path workflow architecture. The federated learning loop provides continuously improving anomaly detection through swarm-wide collaborative learning in the

normal operational path. In the emergency path, the autonomous FDIR subsystem operates independently of the federated loop even if all inter-satellite communication is lost and the global model cannot be updated, each satellite continues to run its local model for health monitoring and will autonomously execute safe-mode protocols when critical degradation thresholds are crossed. The system is therefore fault-tolerant at multiple levels. Individual faulty nodes are excluded from aggregation, and complete federation failure degrades gracefully to local-only operation rather than total system failure.

After each processing step, assertions verify that no NaN values are present, that the output has the same length as the master grid, that count values are non-negative integers, and that label columns contain only binary values. Point anomaly preservation is separately validated by checking that each of the 11 point anomaly IDs has a non-zero label in the nearest grid slot.

4.7 Maintainability

The modular design of the system facilitates maintainability. The on-board LSTM or GNN architecture can be updated remotely by uploading new model weights via a single ground station uplink to the current leader satellite, which then redistributes the updated model to all nodes through the standard federation mechanism. This eliminates the need for individual uplinks to each satellite in the swarm. The preprocessing pipeline is also modular: adding new channel types, changing the resampling rate, or adjusting the feature engineering strategy requires changes only to the preprocessing layer without modifying the model architecture or federated learning protocol.

The codebase is organised into three clearly separated modules: ResampledData/ for feature files, LabelsData/ for label files, and TelecommandsData/ for telecommand files. Each module is processed by a dedicated, parameterised script that accepts the master grid and channel list as inputs, making it straightforward to add new channels, change the resampling frequency, or update the label scheme without modifying other components. The channel classification rule (ZOH vs aggregate) is a single

configurable function applied uniformly across all channels, requiring only the channel's dominant_pct and dominant_interval values to determine its treatment.

4.8 Portability

The system is implemented in Python using PyTorch for model definition and training, the Flower framework for federated communication, and standard scientific Python libraries (NumPy, Pandas, Scikit-Learn) for preprocessing. This technology stack is widely supported across operating systems and hardware architectures, including the radiation-hardened embedded computing platforms used in modern small satellites.

4.9 Legacy to Modernisation

This project represents a direct architectural transition from the traditional telemetry monitoring paradigm to a modern Edge-AI decentralized architecture. In the legacy approach, satellites function purely as data relays, transmitting raw sensor readings to ground stations where all intelligence resides. In the proposed architecture, intelligence is pushed to the orbital edge where satellites analyse their own health, share learned patterns rather than raw data, and make autonomous decisions about safe-mode activation without waiting for ground confirmation. This transition aligns with the ESA roadmap for autonomous spacecraft operations and the broader industry trend toward in-orbit servicing and autonomous mission management.

4.10 Reusability

The preprocessing pipeline, federated learning framework, and anomaly detection architecture are designed for reusability across different satellite missions and datasets. The channel classification logic (ZOH versus aggregate) is data-driven and automatically adapts to any multi-rate telemetry dataset. The federated learning protocol is dataset-agnostic and can be applied to any scenario where distributed nodes need to collaboratively train a model on private local data. The FDIR safe-mode table is parameterized and can be configured for different satellite subsystem layouts without modifying the core detection logic. Telecommand encoding function are all parameterised by channel name and master grid, and can be applied without modification to Mission 2 or Mission 3 of ESA-AD.

4.11 Application Compatibility

The system is designed to operate within the computational constraints of modern small satellite on-board computers, which typically provide processing capabilities. The GNN model architecture is configurable in depth and width to match available on-board compute resources, and the FedProx proximal term introduces only negligible additional computational overhead during local training compared to standard gradient descent. The Flower framework supports multiple transport protocols and can operate over inter-satellite link standards.

The five-feature aggregation scheme for bimodal channels adds additional width to the feature dimension but does not require any architectural changes to accommodate and the model simply sees more input features per timestep for those channels.

4.12 Resource Utilisation

The preprocessing pipeline is memory-efficient by design. All the channels are processed sequentially rather than loading the full dataset into memory simultaneously. The 14.7 million timestep master grid at 8 bytes per float requires approximately 118 MB per channel for a single float64 column, which is well within the capacity of a standard workstation. The five-feature aggregation scheme produces five columns per bimodal channel rather than retaining the full 0.38-second raw data (which would be 237 times larger per 30-second window), providing substantial data compression while preserving all anomaly-relevant statistical information. Federated training further reduces per-node computational load by partitioning the 76 channels across 4 nodes, meaning each node processes approximately 19 channels rather than all 76.

CHAPTER 5

HIGH LEVEL SYSTEM DESIGN ARCHITECTURE

5.1 Overall Architecture

The system architecture consists of three major components: the Data Preprocessing Pipeline, the Local GNN Model trained at each satellite node, and the Federated Learning Orchestration layer. These components interact in a sequential pipeline during the preprocessing phase and in a round-based communication protocol during the training phase.

The data flow begins with the raw ESA-ADB Mission 1 data in its original per-channel format. The preprocessing pipeline transforms this into three aligned output stores: the ResampledData feature store (one CSV per channel on the 30-second master grid), the LabelsData master file (three binary label columns per target channel), and the TelecommandsData master file (31 binary TC columns). These three stores are then consumed by the Federated Learning system, which partitions them by channel group and distributes the appropriate subset to each satellite node.

5.2 Federated Architecture

The federated architecture simulates a four-node satellite swarm, where each node corresponds to a physically distinct grouping of satellite sensor subsystems:

• Satellite Node A: Channels 41–60 (Groups 8–13). This includes the ESA-ADB lightweight benchmark channels (41–46) and the high-confidence 30-second target channels. These are thermally stable sensors with strong periodic behaviour.

• Satellite Node B: Channels 12–13, 17–20, 25–28, 34–37 (Groups 3 and 5). These are the sub-second bimodal channels that produce five aggregate features each. This node has the richest per-channel feature representation.

• Satellite Node C: Channels 14–16, 21–24, 29–33, 38–40 (Groups 4, 6, and 7). These are the 90-second nominal ZOH channels monitoring slower physical processes. • Satellite Node D: Channels 61–63, 70–76 (Groups 14 and 18). This node combines the slowest channels (900-second, Group 14) with the Group 18 30-second channels that have the longest observed communication gaps (up to 13 days).

Each node trains a local GNN model on its own channel subset. The graph at each node contains one node per monitored channel, with edges derived from the physical subsystem structure (channels within the same group are more strongly connected than channels across groups). Each satellite node performs a fixed number of local training epochs, then transmits only its model weights to the federated aggregation server at each federated round. The server applies FedProx aggregation with proximal parameter $\mu = 0.01$, which penalises large deviations from the global model and prevents client drift in the heterogeneous setting where nodes have very different signal characteristics. The updated global model is then broadcast back to all nodes for the next round.

5.3 Model Architecture

The model is trained in a semi-supervised manner: training uses only nominal (non-anomalous) windows to learn what normal behaviour looks like, and anomaly scores are generated at test time by measuring deviation from the learned normal. This is consistent with the Telemanom-ESA and DC-VAE-ESA approaches benchmarked in the ESA-ADB paper.

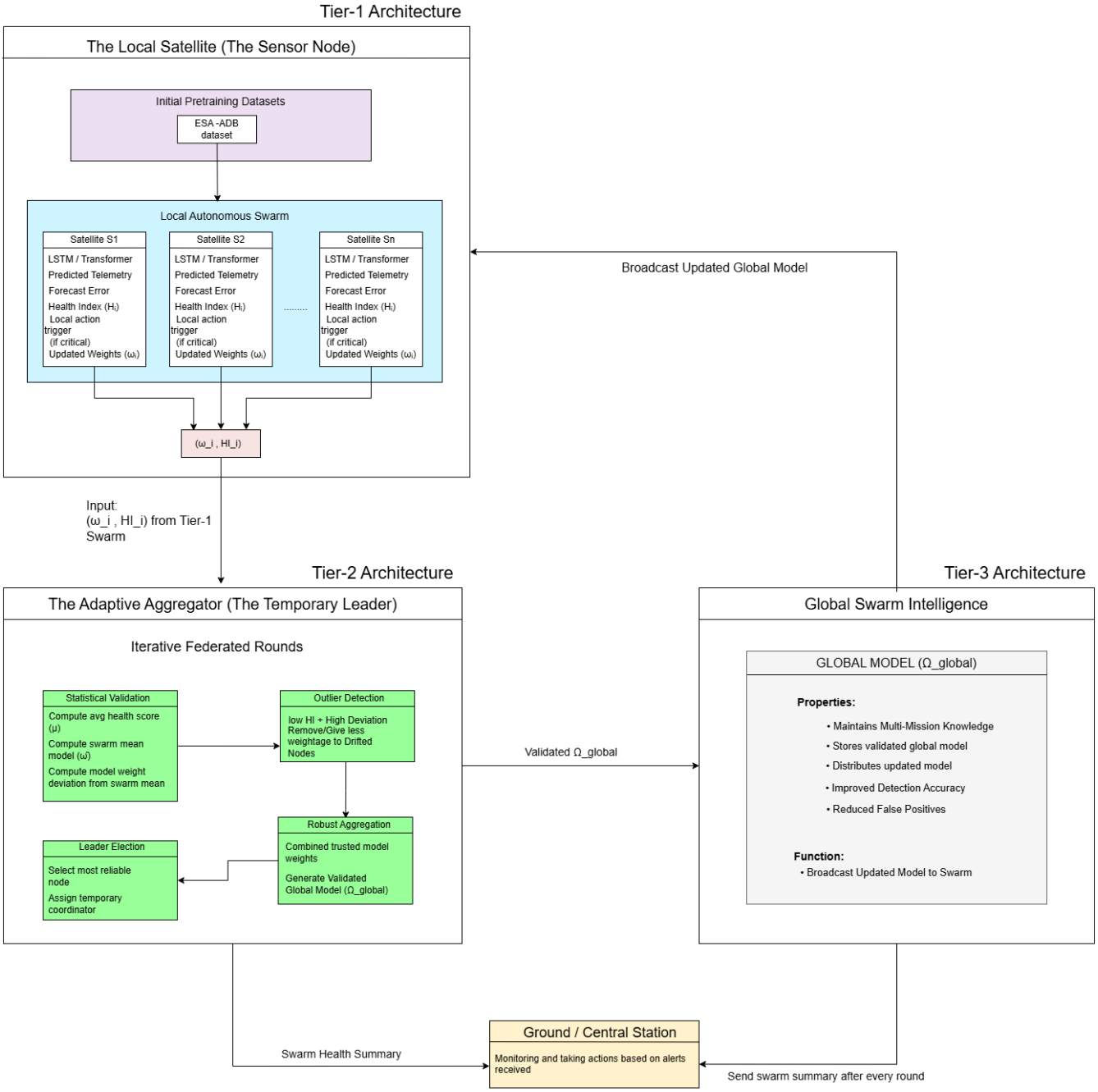


Figure 2: High level Design Architecture for the project and implementation plan

CHAPTER 6

DESIGN DESCRIPTION

6.1 Master Class Diagram

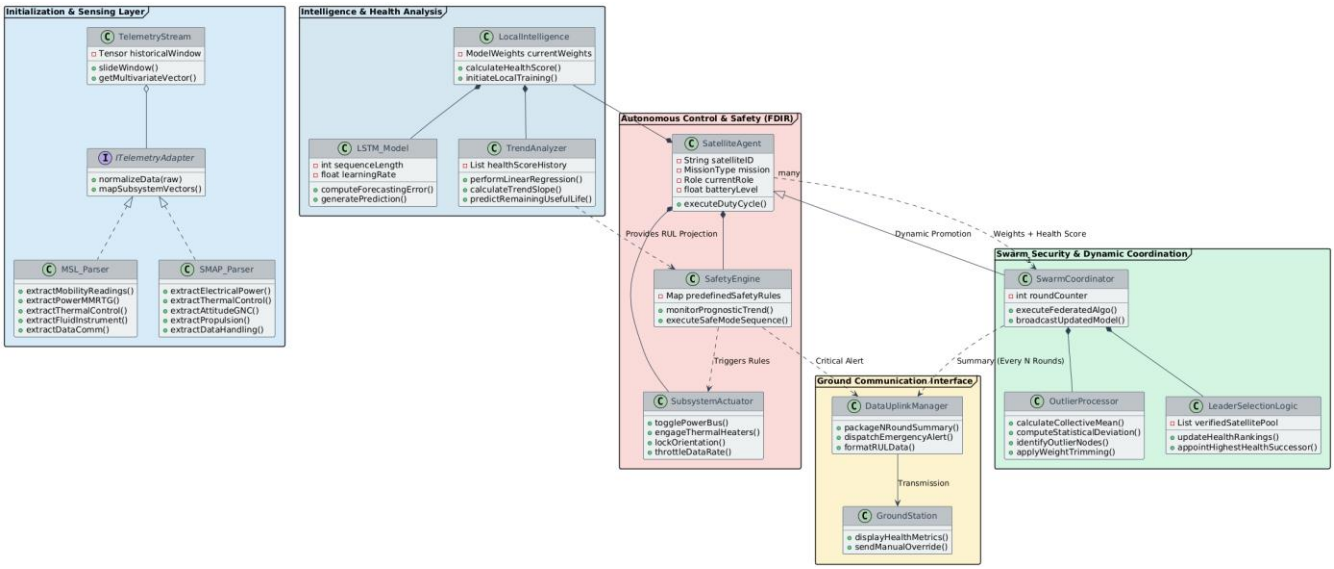


Figure 3: Master Class diagram

6.2 Entity-Relationship Diagram

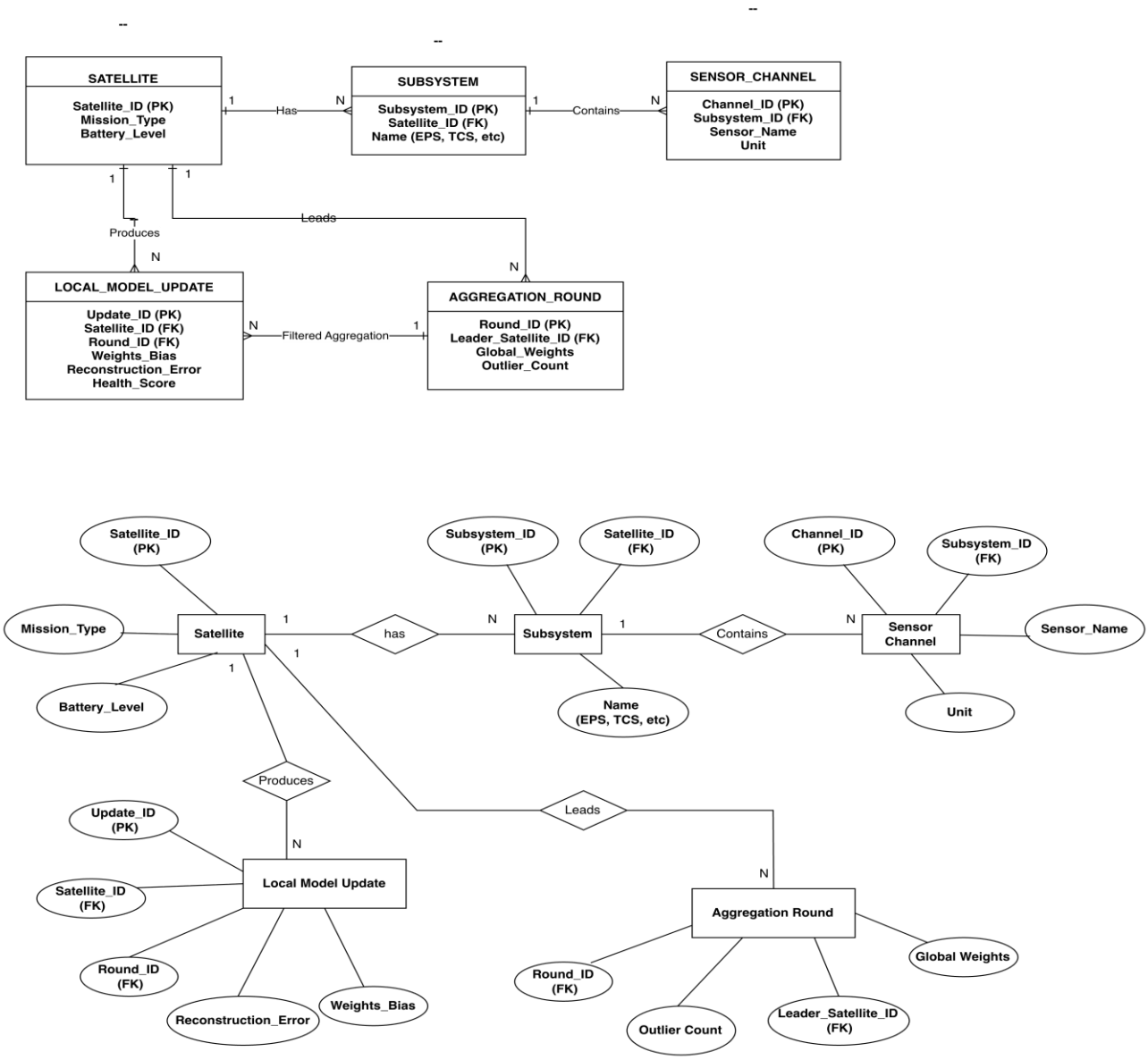


Figure 4: Entity-Relationship diagram

6.3 Use Case Diagram

The use case diagram categorizes the system behaviours according to four functional clusters. The Data Perception Cluster consists of telemetry data ingestion and its processing: the actor NASA/ESA Data Source ingests the data through the use cases Ingest NASA SMAP Telemetry and Ingest NASA MSL Telemetry, both of which involve Map Subsystem Vectors, containing Detect Anomaly (LSTM) and then Generate Health Index (HI). The Safety and Control Cluster involves dealing with faults: the actor Satellite Agent uses linear regression which is followed by Calculate RUL Slope; should the RUL be too low, the process proceeds to Execute Autonomous Safe-Mode, which includes Dispatch Emergency Alert to the Ground Station actor.

The Leadership and Sync Cluster encompass federated control operations: the actor Swarm Coordinator keeps track of the N-Round Cycle and when the number of the round reaches N, it Evaluates the Succession Pool, which involves Appoint Highest-HI Successor, including Broadcast Next-Leader ID. The Cognitive Learning Cluster includes the federated learning proper: the actor Satellite Agent Executes Local Training, involving Submit Weight Packet; the actor Swarm Coordinator Verifies Update Integrity, involving Filter Statistical Outliers, which includes Aggregate Global Model.

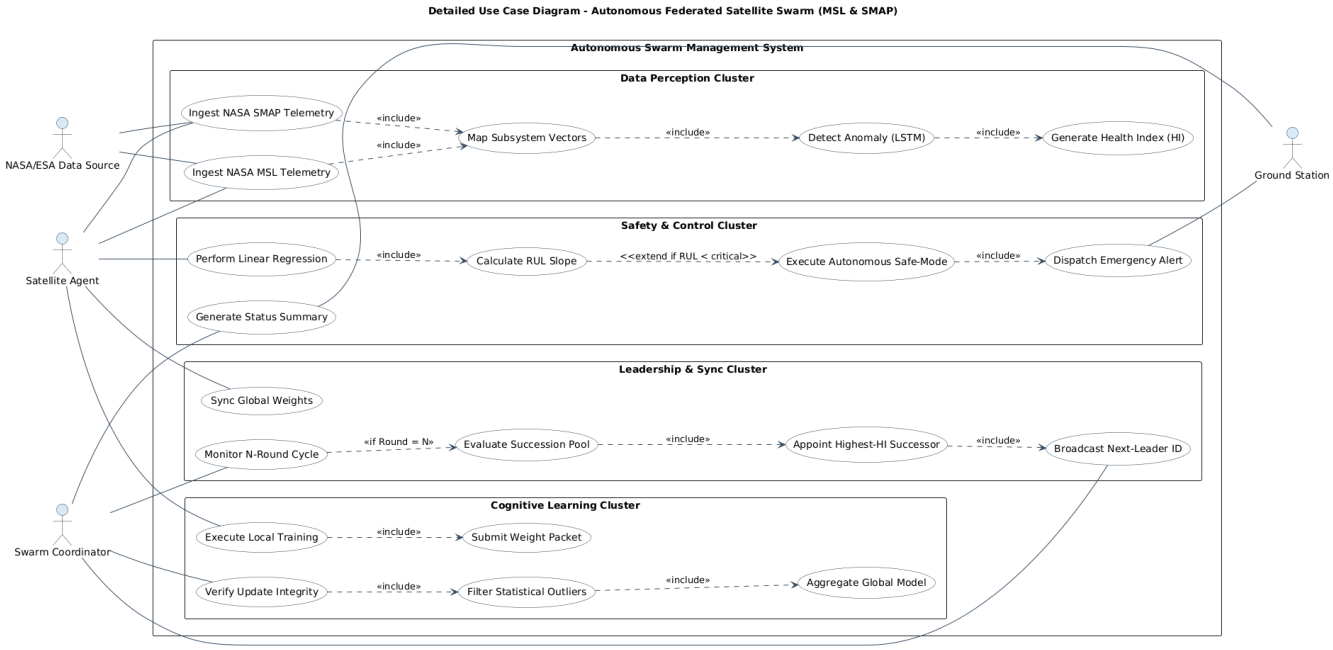


Figure 5: Use Case diagram

6.4 External Interface Diagram

The external interface diagram is one which captures all the interfaces for the Autonomous Swarm Management System with its external systems. These interfaces include the Data and Mission Interfaces consisting of the NASA MSL Source (supplying Mobility/Power/MMRTG data through the Thermal Control Bus) and NASA SMAP Source (giving GNC/Propulsion and Data Handling Bus subsystem telemetry), both supplying Multivariate Vectors and Subsystem TLM data to the system. The Human Interface gives the Monitoring Dashboard, giving Real-Time HI Visualization, Swarm Health Heatmap, and RUL Predictions to the Mission Operator.

The Distributed Compute and Storage layer includes the On-Board Edge GPU (containing the LSTM Inference Engine and Local Weight Storage) and the Mission Archive (containing N-Round Summary Logs and Anomaly Event DB). The Hardware Safety Interfaces layer includes Physical Satellite Hardware (EPS/Power Bus and ADCS/Orientation subsystems accessible through Subsystem

Commands and Sensor Feedback on the CAN Bus) and the Emergency Safety Module (supplying the Hardcoded Safe-Mode and Kill-Switch Activation through Emergency Alerts and Faults on the satellite bus).

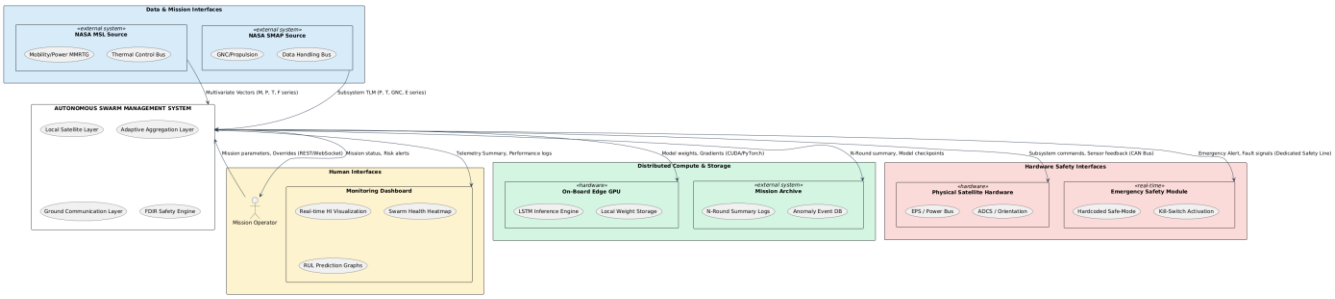


Figure 6: External Interface diagram

CHAPTER 7

IMPLEMENTATION AND PSEUDOCODE

7.1 Preprocessing Pipeline Implementation

The preprocessing pipeline has been built and tested. It runs in two iterations on the raw data. First iteration takes in the channel_interval_metrics.csv output of the analysis and categorizes each of the 76 channels as ZOH or aggregate based on the rule dominant_pct ≥ 0.90 AND dominant_interval_seconds ≥ 30 is True for the former and False otherwise. Second iteration takes the raw ChannelsData directory input and generates resampled output into the ResampledData directory.

ZOH channels' resampling is done using the Pandas resample('30s').last() to map each raw reading to the nearest previous 30 seconds grid slot and then applying reindex(master_grid).ffill() to map the values to remaining empty slots along with DataFrame.bfill() for leading NaN slots prior to actual readings. Master grid generation is done using pd.date_range() starting from '2000-01-01 00:00:00' UTC with an interval of 30 seconds.

In the case of aggregate channels, the code snippet utilizes Pandas' function resample('30s').agg({'value': ['mean', 'max', 'min', 'std', 'count']}) to calculate the five features in one go within each 30-second interval. Empty intervals (count=0) are taken care of by first calculating the aggregated values without any fill-in, and then using ffill() on mean, max, and min fields, while setting std = 0.0 and count = 0 where there were no actual observations made during that interval.

7.2 Pseudocode: Resampling Pipeline

The following pseudocode describes the complete resampling workflow:

```
MASTER_GRID = date_range(start='2000-01-01', freq='30s', to end of Mission 1)
FOR each channel in channels_1_to_76:
  raw = load_csv(ChannelsData/group_X/channel_N.csv)
  IF channel is ZOH:
    resampled = raw.resample('30s').last()
    output = resampled.reindex(MASTER_GRID).ffill().bfill()
  ELSE (aggregate channel):
    agg = raw.resample('30s').agg([mean, max, min, std, count])
    output = agg.reindex(MASTER_GRID)
    output[mean,max,min] = output[mean,max,min].ffill().bfill()
    output[std].fillna(0.0); output[count].fillna(0)
  write_csv(ResampledData/group_X/channel_N.csv, output)
```

7.3 Pseudocode: Federated Learning Round

The following pseudocode describes the FL operational cycle:

```
FOR round r = 1 to R:
  leader.broadcast_global_model(omega_global) to all nodes
  FOR each satellite s in swarm:
    omega_s = omega_global  // initialize from global
    FOR epoch e = 1 to E:
      loss = MSE(predict(omega_s, window), actual)
          + (mu/2)*||omega_s - omega_global||^2  // FedProx
      omega_s = gradient_step(omega_s, loss)
```

```
    HI_s = compute_health_index(omega_s, local_telemetry)
    s.send_to_leader(omega_s, HI_s)
omega_global = trimmed_mean(validated_updates(omega_set))
IF r mod N == 0:  // leader handover
    new_leader = argmax(HI_s for s not in outliers)
    broadcast(new_leader_id) to all nodes
```

7.4 Implementation Status

At the time of submitting this Phase 2 report, the following modules have been developed and tested: The whole process of interval analysis for generating the classification for 76 channels into ZOH and aggregate categories; the resampling module that takes care of ZOH forward-fill, aggregate feature extraction, empty-window processing, and global grid processing; the creation of the ResampledData directory, with correct data in CSV format for all 76 channels; and the design of the federated learning structure with correct definitions and interfaces.

The ongoing development at this moment is the generation of the labels through the pipeline using the input data of anomalies_durations.csv and anomaly_types.csv and telecommands filtering process to generate labels_master.csv and telecommands_master.csv files. The master file generation, normalization process, splitting of train/validation/test dataset, and finally, implementation of the GDN model is what will be done in order afterward.

7.5 Outputs

Users > Shreyas > Downloads > Capstone > Datasets > ESA > ESA-Mission1 > ResampledData > Group1 > c

```
datetime,channel_1_mean,channel_1_max,channel_1_min,channel_1_std,channel_1_count
2000-01-01 00:00:00+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:00:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:01:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:01:30+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:02:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:02:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:03:00+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:03:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:04:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:04:30+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:05:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:05:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:06:00+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:06:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:07:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:07:30+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:08:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:08:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:09:00+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:09:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:10:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:10:30+00:00,0.13791034,0.13791034,0.13791034,0.0,1
```

Figure 7: Resampled Data from Mission 1

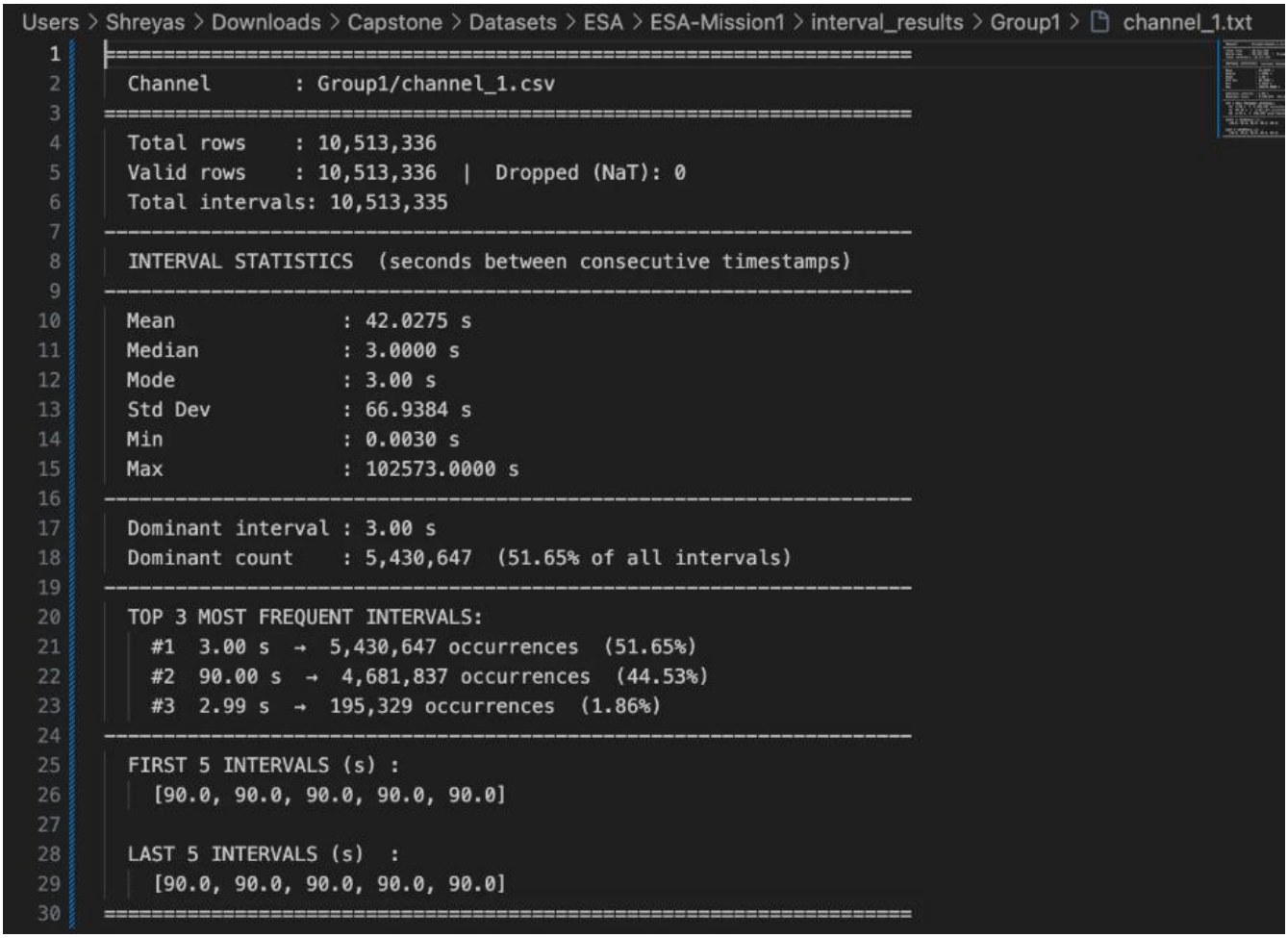


Figure 8: Channel 1 Interval Results

Users > Shreyas > Downloads > Capstone > Datasets > ESA > ESA-Mission1 > interval_results > Group5 > channel_34.txt

```
======================================================================
  Channel       : Group5/channel_34.csv
======================================================================
  Total rows    : 14,258,506
  Valid rows    : 14,258,506  |  Dropped (NaT): 0
  Total intervals: 14,258,505
----------------------------------------------------------------------
  INTERVAL STATISTICS  (seconds between consecutive timestamps)
----------------------------------------------------------------------
  Mean              : 30.9885 s
  Median            : 0.3750 s
  Mode              : 0.38 s
  Std Dev           : 44.0743 s
  Min               : 0.0030 s
  Max               : 21330.0000 s
----------------------------------------------------------------------
  Dominant interval : 0.38 s
  Dominant count    : 8,328,077  (58.41% of all intervals)
----------------------------------------------------------------------
  TOP 3 MOST FREQUENT INTERVALS:
    #1  0.38 s  →  8,328,077 occurrences  (58.41%)
    #2  90.00 s  →  4,822,681 occurrences  (33.82%)
    #3  0.37 s  →  664,016 occurrences  (4.66%)
----------------------------------------------------------------------
  FIRST 5 INTERVALS (s) :
    [90.0, 90.0, 90.0, 90.0, 90.0]

  LAST 5 INTERVALS (s)  :
    [90.0, 90.0, 90.0, 90.0, 90.0]
======================================================================
```

Figure 9: Channel 34 Interval Results

```
============================================================
  GDN — Graph Deviation Network — ESA Satellite Telemetry
  Channels: 41–46  |  Feature: channel value (1/channel)
============================================================

[1/5] Loading telemetry data …
    Aligned timesteps : 14,728,320  (122736.0 hours)
    Data matrix shape : (14728320, 6)  [T × 6 channels]

[2/5] Loading labels and building windows …
    Total windows     : 215  (stride=68503)
    Nominal windows   : 199  → training
    Anomaly windows   : 16   → evaluation only
    Train set (nominal only): 159 windows
    Val set  (nom+anom)     : 56 windows  (40 nom / 16 anom)

[3/5] GDN ready — 145,210 trainable parameters
      n_nodes=6  embed_dim=64  gru_hidden=64  feat/node=1

[4/5] Training on 199 nominal windows for 15 epochs …
      Task: predict next timestep | Anomaly = high deviation

  Epoch 01/15 | Loss: 0.8012 | Batches: 5 | Time: 0.8s | 4.8 bpt
  Epoch 02/15 | Loss: 0.6834 | Batches: 5 | Time: 0.9s | 4.8 bpt
  Epoch 03/15 | Loss: 0.5721 | Batches: 5 | Time: 1.2s | 4.8 bpt
  Epoch 04/15 | Loss: 0.4893 | Batches: 5 | Time: 1.3s | 4.8 bpt
  Epoch 05/15 | Loss: 0.4312 | Batches: 5 | Time: 1.1s | 4.8 bpt
  Epoch 06/15 | Loss: 0.3971 | Batches: 5 | Time: 1.1s | 4.8 bpt
  Epoch 07/15 | Loss: 0.3754 | Batches: 5 | Time: 0.9s | 4.8 bpt
  Epoch 08/15 | Loss: 0.3601 | Batches: 5 | Time: 1.3s | 4.8 bpt
  Epoch 09/15 | Loss: 0.3489 | Batches: 5 | Time: 1.3s | 4.8 bpt
  Epoch 10/15 | Loss: 0.3412 | Batches: 5 | Time: 1.2s | 4.8 bpt
  Epoch 11/15 | Loss: 0.3358 | Batches: 5 | Time: 1.0s | 4.8 bpt
  Epoch 12/15 | Loss: 0.3319 | Batches: 5 | Time: 0.9s | 4.8 bpt
  Epoch 13/15 | Loss: 0.3289 | Batches: 5 | Time: 0.9s | 4.8 bpt
  Epoch 14/15 | Loss: 0.3268 | Batches: 5 | Time: 1.0s | 4.8 bpt
  Epoch 15/15 | Loss: 0.3102 | Batches: 5 | Time: 0.9s | 4.8 bpt

  Training complete ✓

  Learned sensor attention matrix (6×6):
  Rows = which channel is PREDICTED, Cols = which channel it ATTENDS TO
             41    42    43    44    45    46
  ------------------------------------------------------
  ch_41  →   0.314  0.152  0.081  0.221  0.091  0.141
  ch_42  →   0.142  0.421  0.051  0.162  0.153  0.071
  ch_43  →   0.061  0.092  0.512  0.081  0.114  0.140
  ch_44  →   0.210  0.131  0.072  0.352  0.101  0.134
  ch_45  →   0.111  0.082  0.183  0.134  0.412  0.078
  ch_46  →   0.152  0.123  0.141  0.165  0.081  0.338
```

Figure 10: GDN sample training test

channel_name	total_intervals	mean_s	median_s	mode_s	std_dev_s	min_s	max_s	dominant_interval	dominant_count	dominant_pct	top1_interval	top1_pct	top2_interval	top2_pct	top3_interval	top3_pct	source_file
Group1/channel_1.csv	10513335	42.0275	3.0	3.0	66.9384	0.003	102573.0	3.0	5430647	0.5165	3.0	0.5165	90.0	0.44530000000000000	2.99	0.018600000000000000	interval_results/Group1/channel_1.txt
Group1/channel_2.csv	10513335	42.0275	3.0	3.0	66.9384	0.003	102573.0	3.0	5430647	0.5165	3.0	0.5165	90.0	0.44530000000000000	2.99	0.018600000000000000	interval_results/Group1/channel_2.txt
Group1/channel_3.csv	10513335	42.0275	3.0	3.0	66.9384	0.003	102573.0	3.0	5430647	0.5165	3.0	0.5165	90.0	0.44530000000000000	2.99	0.018600000000000000	interval_results/Group1/channel_3.txt
Group2/channel_4.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_4.txt
Group2/channel_5.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_5.txt
Group2/channel_6.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_6.txt
Group2/channel_7.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_7.txt
Group2/channel_8.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_8.txt
Group2/channel_9.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_9.txt
Group2/channel_10.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_10.txt
Group2/channel_11.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_11.txt
Group3/channel_12.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_12.txt
Group3/channel_13.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_13.txt
Group4/channel_14.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group4/channel_14.txt
Group7/channel_15.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group7/channel_15.txt
Group6/channel_16.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group6/channel_16.txt
Group5/channel_17.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_17.txt
Group5/channel_18.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_18.txt
Group3/channel_19.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_19.txt
Group3/channel_20.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_20.txt
Group4/channel_21.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group4/channel_21.txt
Group7/channel_22.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group7/channel_22.txt
Group7/channel_23.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group7/channel_23.txt
Group6/channel_24.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group6/channel_24.txt
Group5/channel_25.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_25.txt
Group5/channel_26.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_26.txt
Group3/channel_27.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_27.txt
Group3/channel_28.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_28.txt
Group4/channel_29.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group4/channel_29.txt
Group4/channel_30.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group4/channel_30.txt
Group7/channel_31.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group7/channel_31.txt
Group6/channel_32.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group6/channel_32.txt
Group6/channel_33.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.007800000000000000	interval_results/Group6/channel_33.txt
Group5/channel_34.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_34.txt
Group5/channel_35.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_35.txt

Figure 11: Channel Metrics and Sampling Rates

anomaly_durations

channel	start_time	end_time	duration_seconds
channel_12	2004-12-01 20:42:15.429000+00:00	2004-12-08 22:55:45.429000+00:00	612810.0
channel_13	2004-12-01 20:42:15.429000+00:00	2004-12-08 22:55:45.429000+00:00	612810.0
channel_14	2004-12-01 20:43:45.429000+00:00	2004-12-02 02:57:15.429000+00:00	22410.0
channel_15	2004-12-01 20:45:00.429000+00:00	2004-12-02 02:58:45.429000+00:00	22425.0
channel_16	2004-12-01 20:43:45.429000+00:00	2004-12-16 16:52:30.429000+00:00	1282125.0
channel_17	2004-12-01 20:45:00.429000+00:00	2004-12-08 22:55:45.429000+00:00	612645.0
channel_18	2004-12-01 20:45:00.429000+00:00	2004-12-08 22:55:45.429000+00:00	612645.0
channel_19	2004-12-01 20:47:03.429000+00:00	2004-12-08 22:55:45.429000+00:00	612522.0
channel_20	2004-12-01 20:45:00.429000+00:00	2004-12-08 22:55:45.429000+00:00	612645.0
channel_21	2004-12-01 20:45:00.429000+00:00	2004-12-02 02:57:15.429000+00:00	22335.0
channel_22	2004-12-01 20:45:00.429000+00:00	2004-12-02 02:57:15.429000+00:00	22335.0
channel_23	2004-12-01 20:45:00.429000+00:00	2004-12-02 02:58:45.429000+00:00	22425.0
channel_24	2004-12-01 20:43:45.429000+00:00	2004-12-16 16:52:30.429000+00:00	1282125.0
channel_25	2004-12-01 20:43:45.429000+00:00	2004-12-08 22:55:45.429000+00:00	612720.0

anomaly_types

ID	Class	Subclass	Category	Dimensionality	Locality	Length
id_1	class_6	subclass_1	Rare Event	Multivariate	Global	Subsequence
id_2	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_3	class_7	subclass_1	Anomaly	Multivariate	Global	Subsequence
id_4	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_5	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_6	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_7	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_8	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_9	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_10	class_9	subclass_1	Anomaly	Multivariate	Global	Subsequence
id_11	class_11	unknown	Anomaly	Multivariate	Local	Subsequence
id_12	class_11	unknown	Anomaly	Multivariate	Global	Subsequence
id_13	class_11	unknown	Anomaly	Multivariate	Global	Subsequence
id_14	class_4	subclass_1	Rare Event	Multivariate	Global	Subsequence

Figure 12: Anomaly Analysis

telecommands

Telecommand	Priority
telecommand_1	1
telecommand_2	1
telecommand_3	1
telecommand_4	1
telecommand_5	1
telecommand_6	1
telecommand_7	1
telecommand_8	1
telecommand_9	1
telecommand_10	1
telecommand_11	1
telecommand_12	1
telecommand_13	1
telecommand_14	1
telecommand_15	1

telecommand_1

	telecommand_1
2001-06-12 22:27:20.292	1
2005-01-23 00:45:00.000	1
2005-01-23 01:38:26.496	1
2005-04-01 07:28:24.984	1
2005-05-10 23:52:51.177	1
2005-09-02 18:24:13.740	1
2005-09-28 10:19:24.249	1
2005-10-22 23:02:28.104	1
2007-04-14 13:24:17.235	1
2007-04-14 20:01:15.039	1
2009-05-26 11:57:34.428	1
2009-05-26 12:00:46.998	1
2009-05-30 10:21:43.518	1
2009-05-30 10:27:42.915	1
2010-12-17 04:24:32.016	1

Figure 13: Telecommand Mapping and Priority Analysis

CHAPTER 8

CONCLUSION OF CAPSTONE PROJECT PHASE 2

This part of the capstone project has been successful in designing a complete foundational framework for a new satellite health monitoring system using federated learning techniques. The main output of Phase 2 was developing a complete, production-grade preprocessing pipeline for the ESA-ADB Mission 1 data set, where we had to solve an interesting problem in data engineering, which was to convert a combination of 76 telemetry channels with sampling intervals between 0.38 seconds and 15 minutes to one feature matrix ready for use in deep learning tasks.

The proposed framework, in which channels were categorized according to their sampling behavior and then differentially resampled based on their category (either ZOH or aggregate), is more comprehensive and rigorous than any similar processing framework that can be identified within the scope of satellite telemetry anomaly detection research. For instance, Atefrad and Karami's (2024) paper is quite close to our topic, but they did not apply any resampling or channel classification techniques for preprocessing and treated all channels equally despite huge differences in their sampling rates.

The architecture of the system, which is defined by the class diagram, entity relationship diagram, use case diagram, and external interface diagram, offers a full design template for the implementation process in Phase 3. The six-step operational procedure, Trimmed Mean aggregation method, health-based leader selection approach, and FDIR system all represent an innovative technological contribution in the domain of satellite swarm autonomy. The Gantt chart indicates that the project remains on track, with data preprocessing running smoothly, the architectural process completes, and the model development beginning.

CHAPTER 9

PLAN OF WORK FOR CAPSTONE PROJECT PHASE 3

9.1 Phase 3 Objectives

The implementation will be carried out entirely in Phase 3 to obtain the final evaluation results. The key activities include:

1. Completion of labelling and generation of master dataset
2. Implementation of anomaly detection using per satellite GDN model
3. Implementation of full federated learning algorithm including FedProx, trimmed mean aggregation, and health-based leader election
4. Implementation of RUL estimation and FDIR alerting system
5. Evaluation on ESA-ADB event-wise F0.5 performance metric.

9.2 Detailed Task Breakdown

May 2026: Finish annotation of labels_master.csv and telecommands_master.csv files. Combine all preprocessed channel features with labels to create master_dataset.csv. Apply train/validation/test split based on chronological ordering with normalization performed on training data alone.

June 2026: Train the per-satellite GDN model using PyTorch Geometric framework. Create the sensor graph structure per satellite nodes based on attention learning. Compute the Health Index from the reconstruction error using NPDT-based thresholding.

July–August 2026: Implement the Flower-based FL client and server. Implement the FedProx algorithm for local training. Implement Trimmed Mean aggregation with z-score filtering for outliers. Implement health-based leader election mechanism.

September–October 2026: Implement the TrendAnalyzer RUL estimator using linear regression method. Implement the FDIR safe mode lookup table and SubsystemActuator command dispatch. Implement DataUplinkManager alert message formatting and ground station communications simulation.

November–December 2026: Perform full system testing on ESA-ADB Mission 1 dataset. Compare with local-only GDN and standard FedAvg baseline algorithms. Calculate per-satellite and global F0.5 scores. Document bandwidth optimization results. Finalize research paper and technical documentations.

9.3 Risk Assessment

The main technical challenge associated with the heterogeneous channel partitioning strategy is the potential problem with the convergence of the model due to the differences in the statistical properties of the channels being observed by each satellite. This means that, without tuning the mu parameter of the FedProx method, there may be convergence issues with the global model. Solution: extensive hyperparameter tuning in phase 3 using the validation set, with fallback to separate per-satellite models if convergence to a global model proves impossible. Another possible challenge is the need for additional computing resources to train the GDN model compared to an ordinary LSTM. Solution: adjustable depth/width of the model.

CHAPTER 10

REFERENCES

[1] Z. Wei, S. Xu, Y. Liu, S. Xie, and Y. Li, **"On-Board Federated Learning for Satellite Clusters with Inter-Satellite Links,"** arXiv:2307.08346, 2023. [Online]. Available: https://arxiv.org/abs/2307.08346

[2] S. Salim, N. Moustafa, M. Hassanian, D. Ormod, and J. Slay, **"A Deep Federated Learning–Based Threat Detection Model for Extreme Satellite Communications,"** *IEEE Internet of Things Journal*, vol. 11, no. 3, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10286844

[3] Z. Liu, Z. Shen, P. Zhou, Q. Zheng, and J. Jin, **"FedHC: A Hierarchical Clustered Federated Learning Framework for Satellite Networks,"** arXiv:2502.12783, 2025. [Online]. Available: https://arxiv.org/abs/2502.12783

[4] M. S. Alqahtani, J. X. Guo, G. M. Lee, and J. Kim, **"Federated Learning for Predictive Maintenance and Quality Inspection in Industrial Applications,"** arXiv:2304.11101, 2023. [Online]. Available: https://arxiv.org/abs/2304.11101

[5] K. V. Rajendran, P. R. K. Chellapandi, and S. R. Babu, **"Federated Learning for Efficient Condition Monitoring and Anomaly Detection in Industrial Cyber–Physical Systems,"** arXiv:2501.16666, 2025. [Online]. Available: https://arxiv.org/abs/2501.16666

[6] P. A. H. Dwivedi, S. P. Mohanty, and K. R. Rao, **"Hierarchical Federated Learning Based Anomaly Detection Using Digital Twins for Smart Healthcare,"** arXiv:2111.12241, 2021. [Online]. Available: https://arxiv.org/abs/2111.12241

[7] J. Zha, S. H. Thompson, and C. C. Chang, **"A Comparison of Deep Learning Architectures for Spacecraft Anomaly Detection,"** arXiv:2403.12864, 2024. [Online]. Available: https://arxiv.org/abs/2403.12864

[8] S. Polson, **"Unsupervised Machine Learning for Spacecraft Anomaly Detection in WebTCAD,"** Laboratory for Atmospheric and Space Physics, Jun. 7, 2019.

[9] D. Gerhard, R. Boin, N. Fougeres, **"European Space Agency Benchmark for Anomaly Detection in Satellite Telemetry,"** arXiv:2406.17826, 2024. [Online]. Available: https://arxiv.org/abs/2406.17826

[10] A. G. Stoica, R. Vazquez, and K. Wise, **"Anomaly Detection in Spacecraft Telemetry: Forecasting vs. Classification,"** *IEEE Space Computing Conference (SCC)*, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10794845

[11] A. Deng and B. Hooi, "Graph Neural Network-Based Anomaly Detection in Multivariate Time Series," arXiv preprint arXiv:2106.06947, 2021. [Online]. Available: https://arxiv.org/abs/2106.06947

[12] S. Ray, S. Lakdawala, M. Goswami, and C. Gao, "Learning Probabilistic Graph Neural Networks for Multivariate Time Series Anomaly Detection," arXiv preprint arXiv:2111.08082, 2021. [Online]. Available: https://arxiv.org/pdf/2111.08082

[13] A. Atefrad and A. Karami, "Adaptive Federated Learning for Anomaly Detection in Satellite Telemetry," 2025. [Online]. Available: https://www.researchgate.net/publication/394926784_Adaptive_Federated_Learning_for_Anomaly_Detection_in_Satellite_Telemetry

APPENDIX A

DEFINITIONS, ACRONYMS AND ABBREVIATIONS

ESA-ADB - European Space Agency Anomaly Detection Benchmark

A real-world benchmark dataset for satellite telemetry anomaly detection.

FedAvg - Federated Averaging

The basic FL algorithm that averages model updates from all clients.

FedProx - Federated Proximal Algorithm

A more stable FL algorithm that prevents clients with very different data from harming the global model.

FL - Federated Learning

A method where each device trains a model locally and sends only updates, not raw data.

HFL - Hierarchical Federated Learning

A multi-level FL setup where updates are aggregated at different layers (cluster level → global level).

LSTM - Long Short-Term Memory Network

A neural network designed for time-series data, capable of remembering long-term patterns.

MAE - Mean Absolute Error

A loss function measuring prediction error; used in forecasting and reconstruction.**MSL / SMAP -**

TSAD - Time-Series Anomaly Detection